

Leave Clear Text Alone: Does an Editing Skill Keep Coding Agents from Making Open-Source Prose Worse?

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<https://github.com/ting-hong-shieh/polish-open-source-prose>

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Abstract

Coding agents asked to polish open-source prose tend to rewrite it. We test whether a small Agent Skill, *polish-open-source-prose* v0.2.0, keeps them from making text worse, and whether it does more than a short instruction. Seven model configurations from Anthropic and OpenAI ran under five conditions (no instruction, a one-sentence and a three-sentence instruction, and two versions of the skill) on 49 development cases, 24 held-out cases written after release, and 21 real pull request descriptions that reviewers at Apache Airflow, Arrow, DataFusion, and CPython had criticized. Literal checks scored every output, and two judge models graded the held-out outputs blind. Without an instruction the models changed 96–98% of texts that should have been left alone, and the judges rated 30–44% of all such outputs harmful. A three-sentence instruction recovered most of the restraint (81–85% left unchanged, against 88–92% with the skill; the difference is not significant), but it also left unsupported claims in place. On held-out cases the skill fixed them cleanly about twice as often (0.71–0.76 against 0.34). On the real pull requests no condition cleanly fixed more than a quarter of the reviewers’ critiques. Without an instruction, 21–38% of rewrites added facts and 23–27% damaged needed text, including other authors’ AI disclosure lines; with the skill these fell to 0–1% and 1–4%, though it fixed fewer problems than its longer first version. The skill costs 2.5 to 3.3 times as much per run in Claude Code. The cases, scores, judge labels, and scripts are in the project repository; the real pull request text is not.

1 Introduction

Contributors now ask coding agents to write and tidy the prose around their code: pull request descriptions, issue replies, commit messages, documentation, and code comments. Several large open-source projects have responded with explicit rules. Reviewers in those projects ask for evidence behind claims, for descriptions that match the change, and for disclosure of AI use (Section 2).

An agent asked to “polish” a passage tends to treat the request as an obligation to change it. That creates two risks that matter more to a reviewer than style does. The agent may rewrite text that was already clear, which costs review time and can shift meaning; or it may make the text sound more finished than the work behind it, for example by turning “probably” into “clearly” or adding a test result nobody ran.

This report evaluates *polish-open-source-prose*, a small Agent Skill written to prevent those two failures. Version 0.2.0 has four rules and a final check: leave clear text alone, keep exact elements exact, do not add facts, and edit only where there is a concrete cost. We ask:

- RQ1** How often do current models change text that should be left alone, and how much does the skill reduce that?
- RQ2** Does the skill do more than a short instruction that states the same intent in one or three sentences?
- RQ3** Does the restraint cost anything on text that does need editing?
- RQ4** Do the results hold on cases written after the skill was frozen and on real pull request descriptions that reviewers criticized?
- RQ5** What does the skill cost in time and tokens?

We ran seven model configurations from two vendors under five conditions on three case sets, scored every output with literal checks, and had two judge models grade the held-out outputs blind. The cases, scores, judge labels, and scripts are in the repository; the text of the real pull requests is not, because it belongs to their authors (Section 5).

2 Background

Agent Skills. A skill is a folder with a `SKILL.md` file whose frontmatter description tells the agent when to load it, plus optional reference files the agent reads on demand. Claude Code and Codex both support the format. Only the description sits in the context until the skill is used, so a skill’s cost is paid on the tasks that load it.

Project policies on AI-assisted contributions. The Apache Software Foundation’s generative tooling guidance¹ allows AI-assisted contributions under conditions and recommends that commit messages carry a `Generated-by:` token. The Rust project adopted an LLM usage policy² that does not accept pull request descriptions or comments that originate from an LLM, while allowing some uses, such as private review and translation, with disclosure. Apache Airflow added review instructions aimed at low-effort AI-generated contributions,³ and the DataFusion contributor guide has a section on AI-assisted contributions.⁴ A public collection of such policies is maintained on GitHub.⁵

None of these policies asks for prose that sounds less machine-written. They ask for claims that can be checked, descriptions that match the change, and disclosure. That shaped the skill’s scope: version 0.1.0 had grown from a list of phrases typical of model prose (it drew on stop-slop⁶), and version 0.2.0 dropped the phrase lists in favor of rules about fidelity.

What reviewers object to. To see which problems reviewers raise, we searched pull request comments in `apache/airflow`, `apache/datafusion`, `apache/arrow`, and `python/cpython` from 2025-10-01 to 2026-10-04 for phrases such as “AI-generated”, “LLM”, and “provide evidence”, and saved 459 matching threads with their description edit history and title changes. A stratified sample of 48 threads was labeled by a Claude subagent and spot-checked. Figure 1 shows the critique types. Thirty-three of the 48 threads criticized the text itself, not only the code. The most common complaints were claims without evidence (24 threads) and undisclosed AI use (19). Padded or hard-to-read text came up less often (7 and 6 threads). Authors edited the description or title

¹<https://www.apache.org/legal/generative-tooling.html>

²<https://forge.rust-lang.org/policies/llm-usage.html>

³<https://github.com/apache/airflow/pull/62442>

⁴<https://datafusion.apache.org/contributor-guide/index.html#ai-assisted-contributions>

⁵<https://github.com/melissawm/open-source-ai-contribution-policies>

⁶<https://github.com/hardikpandya/stop-slop>

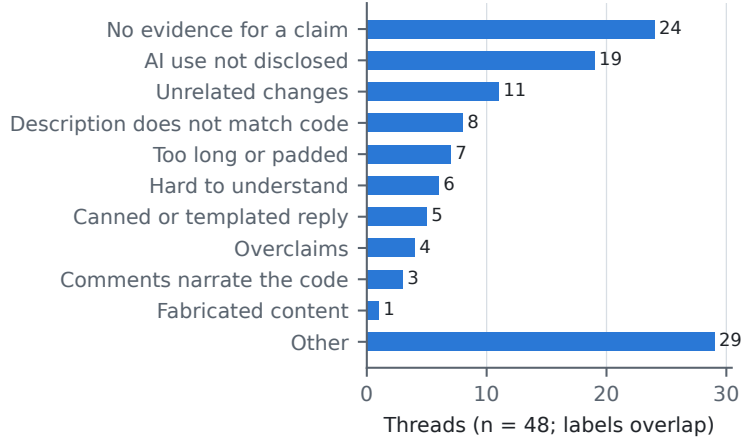


Figure 1: Critique types in 48 labeled review threads from four projects. The sample comes from a phrase search, so it shows what reviewers object to, not how often.

afterwards in 13 threads. The label “Other” (29) covers critiques of the code, the process, or the contributor’s conduct; a thread can carry several labels.

3 Related work

Over-editing. Studies of grammatical error correction found that ChatGPT tends to over-correct, producing fluent rewrites where a minimal edit was expected [2]. Our keep cases measure the same tendency in a coding-agent setting, where the cost is not a lower correction score but a reviewer reading changes nobody asked for.

Instruction following. IFEval scores whether a model follows instructions that can be checked mechanically, such as length or required keywords [9]. Our literal checks follow the same idea for an editing task: strings that must survive, must go, or must not appear, plus exact equality for text that should not change.

LLM judges. Using strong models as judges agrees reasonably well with human preferences but shows position, verbosity, and self-enhancement biases [8], and models can recognize and favor their own outputs [5]. We therefore hide model and condition from the judges, use two judges from different vendors, report them separately, and measure self-preference (Section 6.5).

Detection and provenance. Detectors of model-written text misclassify writing by non-native English speakers as machine-generated [4] and can be defeated by paraphrasing [6]. Watermarks such as SynthID-Text are applied while the model samples tokens [1], so a later editing step cannot add or certify one. For these reasons the skill does not try to make text sound human or pass a detector, and v0.2.0 dropped the phrase lists that v0.1.0 had used for that purpose.

Agent Skills. Anthropic introduced the skill format with progressive disclosure: the agent sees each skill’s name and description and loads the body when a task needs it [7]. A study of 238 SKILL.md files found that over 99% violate at least one authoring best practice [3]. That work

examines how skills are written; this report measures what one skill changes in agent output, against instructions short enough to put in a prompt.

4 What we compared

Conditions. Every case was run under five conditions. The task prompt was the same in all of them; only the added instruction differed.

- **No skill:** the task prompt alone.
- **One sentence:** “If the text is already clear and fits its surface, returning it unchanged is a valid answer.”
- **Three sentences**⁷: “Polish this text only where there is a concrete problem. Leave already-clear text unchanged. Preserve facts, scope, conditions, negation, commands, links, quotations, and author voice. Do not invent missing facts.”
- **Skill v0.1.0:** the first release, a style guide of 1,175 words in `SKILL.md` plus six reference files (about 3,900 words) with English and Chinese pattern lists, surface notes, and examples.
- **Skill v0.2.0:** the current release, 943 words in `SKILL.md` plus two reference files (a Taiwan locale pack and a provenance note, 606 words together).

The short instructions were prepended to the task prompt. They were written once, in English, and not tuned.

Models and harnesses. Seven configurations: Claude Opus 5.5, Opus 4.6, and Sonnet 5.5 at high effort in Claude Code; GPT-6.1 Sol at high, GPT-6 Astra at high and low, and GPT-6 Luna at max reasoning effort in Codex CLI 0.160.0. Claude Code ran with only the Read and Skill tools, no MCP servers, and no session persistence; skill arms forced the skill with a slash command naming a project-local copy of the version under test. Codex ran in a `bwrap` sandbox with an empty home directory; skill arms placed the skill under `.agents/skills` with an `AGENTS.md` line asking the agent to use it. Event logs show that every Codex skill run opened `SKILL.md`.

Claude configurations ran each case twice per condition and we report the mean of the two runs; Codex configurations ran once. The two harnesses use different task prompts. Claude received the task in the case’s language and was asked to put the revision inside `<revised>` tags. Codex received the same one-line Chinese task prompt for every case (inherited from the first experiment), asking for the usable result only, with the surface and target locale filled in. We therefore compare conditions within a configuration, not vendors.

5 Benchmark

Development set (49 cases). The skill’s own forward cases at commit `2a05713`: 19 *keep* cases whose correct answer is the input unchanged, 26 *rewrite* cases, 2 *audit* cases, and 2 provenance questions (left out of the scores below). Twenty-one are in English and 28 in Traditional Chinese (Taiwan). The skill was developed against these cases, and seven expected answers were corrected during development, so this set measures fit, not generalization.

Held-out set A1 (24 cases). Written after v0.2.0 was released and frozen at commit `fa993da` before any model ran on it: 8 *keep* and 8 *rewrite* cases in English, 4 and 4 in Traditional Chinese. Each case follows the pattern of a real reviewer critique (linked in the case) but uses new wording

⁷The instruction has four sentences; the name stuck from an earlier draft.

and invented project details, and each has a short rubric. The cases were drafted with Claude Opus 5.5, which is also one of the tested models.

Held-out set A2 (21 cases). Real pull request titles and descriptions from apache/airflow (11), apache/arrow (6), apache/datafusion (2), and python/cpython (2), as they stood before a reviewer criticized their text, selected from the collected threads. Each case keeps the reviewer’s critique, a description of what a careful editor would do, and the author’s later revision for scoring; the model sees only the original. Critique kinds: overclaim (6), description does not match the change (4), misleading title (4), unclear (3), AI-sounding (2), other (2). In 12 of the 21 cases the critique depends on something outside the text, such as the diff or CI results, so no text-only editor can fully address it; the best an editor can do there is not make things worse and ask. The text stays outside Git; `evals/heldout/a2_manifest.json` records the URLs and a SHA-256 of the case file, committed before any run.

Literal scoring. Each development and A1 case lists strings that must be preserved, must be removed, or must not be introduced. Matching is on NFC-normalized, case-sensitive substrings. A keep case passes only if the output equals the input, apart from trailing whitespace.

Judged scoring. Two judges, Claude Opus 5.5 and GPT-6.1 Sol at high effort and without the skill, graded outputs blind: all outputs for one case were shuffled under random labels and graded in groups of up to 20. For A1 and A2 edit cases the judge saw the original, the rubric (A1) or the reviewer’s critique and ideal behavior (A2), and each output’s revised text. It decided whether the output addressed the problem (0, 0.5, or 1), invented a fact, or damaged something needed. Claude Code outputs usually carried a note to the author outside the `<revised>` tags, such as a question about where a number came from. The judge saw that note too: a question in it counted toward addressing a problem the text alone cannot fix, but only the revised text counted for invented or damaged facts. The Codex prompt asked for the usable result only, so Codex outputs had no such channel. For keep cases in the development set and A1, every output that changed the text was judged for whether the change was harmful: whether it altered meaning, a fact, a condition, scope, an obligation, a quotation, an identifier, or the author’s voice, or added information. Both judges are also tested models, so we report them separately and check for self-preference (Section 6.5).

6 Results

Tables 1 and 2 give literal pass rates for every configuration and condition. Figure 4 pools the seven configurations, giving each the same weight, with 95% intervals from resampling cases.

6.1 RQ1: Models rewrite clear text unless told not to

With no added instruction, every configuration changed almost every keep case: 0–16% were returned unchanged on the development set and 0–12% on A1 (Figure 2). Many of those changes were not harmless rewording. The judges rated 39–44% of all no-skill outputs on development keep cases, and 30–38% on A1, as changing meaning, a fact, a condition, an obligation, an identifier, or the author’s voice (Figure 3). Changes that both judges rated harmful include translating an English license notice into Chinese, adding a confirmation step to an ordered procedure, adding a disclosure condition to a security policy, replacing a colloquial phrase that was the author’s voice, and turning “tested” into “verified”.

Table 1: Development set: percentage of cases passing, by model and condition (mean over runs; 19 keep and 28 edit cases; the 2 provenance cases are left out).

	No skill		One sentence		Three sentences		Skill v0.1.0		Skill v0.2.0	
	keep	edit	keep	edit	keep	edit	keep	edit	keep	edit
Opus 5.5	8	73	68	79	82	64	71	75	100	89
Opus 4.6	8	68	76	68	82	43	50	88	95	84
Sonnet 5.5	16	68	79	71	92	61	92	82	100	89
GPT-6.1 Sol	0	64	42	54	79	43	32	82	95	89
GPT-6 Astra (high)	0	57	42	64	90	50	26	75	100	93
GPT-6 Astra (low)	0	61	16	57	79	43	10	71	100	93
GPT-6 Luna (max)	0	36	37	39	63	39	5	57	53	68

Table 2: Held-out set A1: percentage of cases passing literal checks (12 keep, 12 edit).

	No skill		One sentence		Three sentences		Skill v0.1.0		Skill v0.2.0	
	keep	edit	keep	edit	keep	edit	keep	edit	keep	edit
Opus 5.5	0	33	71	42	79	62	79	75	96	83
Opus 4.6	12	38	58	58	79	50	54	54	92	71
Sonnet 5.5	4	46	92	58	92	58	88	75	92	83
GPT-6.1 Sol	0	42	25	25	83	25	0	50	92	75
GPT-6 Astra (high)	0	33	33	42	92	42	8	42	92	92
GPT-6 Astra (low)	0	50	8	33	100	25	0	42	92	92
GPT-6 Luna (max)	0	33	33	33	67	17	0	25	58	58

With v0.2.0, six of the seven configurations returned 92–100% of keep cases unchanged on both sets. GPT-6 Luna at max effort was the exception, at 53% and 58%: it still translated license text, replaced the UI label, and reworded commit references.

6.2 RQ2: A three-sentence instruction recovers most of the restraint

Pooled over configurations, the three-sentence instruction left 81% of development keep cases unchanged (95% interval 69–91%) and 85% of A1 keep cases (69–96%), against 92% (87–95%) and 88% (72–98%) for v0.2.0. The difference is not significant: v0.2.0 did better on 8 development cases and worse on 2 ($p = 0.11$), and better on 2 A1 cases and worse on 3 ($p = 1$; Table 3). The judged harm rates are low for both: 4–5% for the three-sentence instruction and 2% for v0.2.0 on the development set, and at most 4% and 1% on A1.

The one-sentence instruction was much weaker (52% and 46% unchanged). Version 0.1.0, the longer style guide, was weaker still on this measure (41% and 33%): for the four GPT configurations it left fewer keep cases unchanged than the one-sentence instruction did. A style guide full of patterns to fix plausibly gives the model more reasons to edit.

6.3 RQ3: The short instruction gives up the edits that matter

The three-sentence instruction pays for its restraint on text that does need work. On edit cases it passed 49% of literal checks on the development set and 40% on A1, no better than no instruction (61% and 39%), while v0.2.0 passed 86% and 79% (Figure 4). Against the three-sentence instruction, v0.2.0 did better on 20 development edit cases and worse on 1 ($p < 0.001$), and better on 9 A1 edit

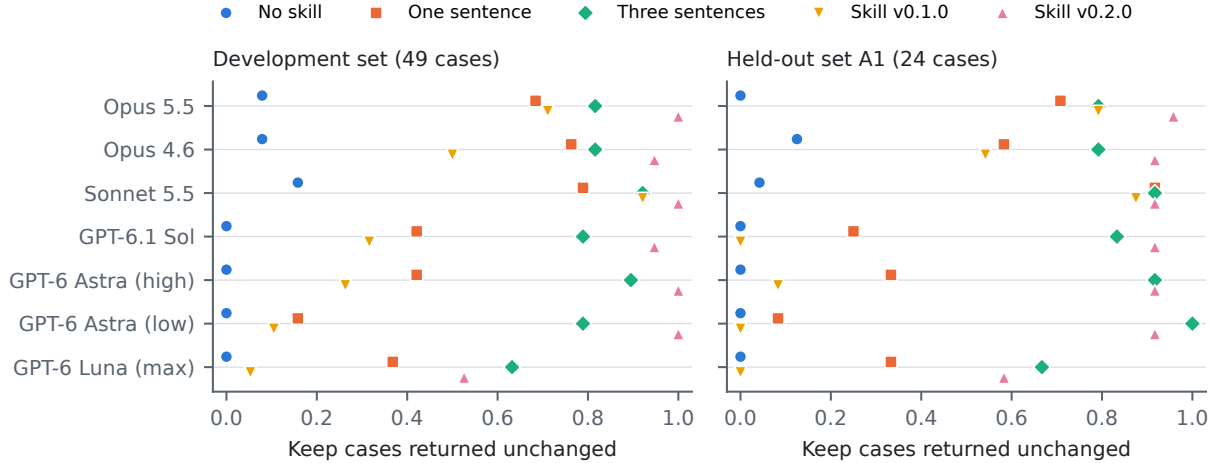


Figure 2: Share of keep cases returned unchanged, by configuration and condition.

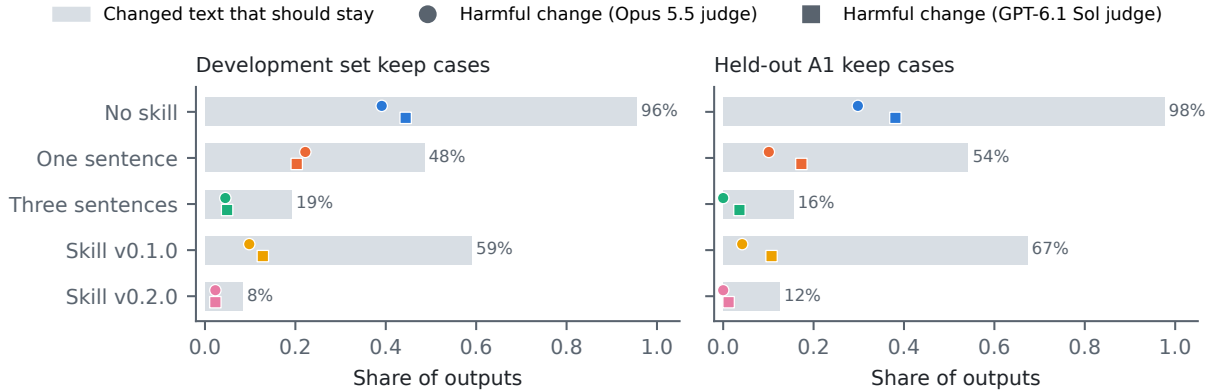


Figure 3: Keep cases: the bar is the share of outputs that changed the text; the markers are the share of all outputs whose change a judge rated harmful.

cases and worse on 1 ($p = 0.02$).

The judges explain why (Table 4). Under the three-sentence instruction the models returned 56% of A1 edit cases unchanged, leaving claims such as “3x faster” with no benchmark, or “fully tested and safe to merge” after one test file, in place. In some of these the model’s own note named the problem and kept the text anyway, because the instruction said to leave clear text unchanged (Appendix B). Its mean score for addressing the problem was 0.34 from both judges, against 0.76–0.77 for v0.2.0. Without any instruction the models addressed more problems (0.60–0.61), but 11–19% of their outputs added a fact the author had not given, against 1% for v0.2.0. Counting a fix only when the revision neither added a fact nor damaged needed text (a *clean fix*), v0.2.0 scored 0.71–0.76, the three-sentence instruction 0.34, and no instruction 0.45–0.49; v0.2.0 did better than both on most cases (12:0 and 10:1; Table 3). Version 0.1.0 came close (0.67–0.71; 9:3, $p = 0.15$). The skill differs from the short instruction in naming what counts as a problem: unsupported certainty, quality claims without evidence, and test results stretched beyond their scope.



Figure 4: Keep cases left unchanged against edit cases passing literal checks, pooled over seven configurations. Bars are 95% intervals from resampling cases.

Table 3: Paired sign tests of v0.2.0 against each other condition: cases where v0.2.0 scored better:worse (two-sided p). Each case’s score is averaged over the seven configurations, their runs, and, for judged scores, both judges; ties are dropped.

	No skill	One sentence	Three sentences	Skill v0.1.0
Development, keep (literal)	19:0 (<.001)	17:0 (<.001)	8:2 (.109)	19:0 (<.001)
Development, edit (literal)	17:4 (.007)	19:2 (<.001)	20:1 (<.001)	13:4 (.049)
A1, keep (literal)	12:0 (<.001)	11:0 (<.001)	2:3 (1.000)	12:0 (<.001)
A1, edit (literal)	11:1 (.006)	11:0 (<.001)	9:1 (.021)	9:0 (.004)
A1, edit (judged clean fix)	10:1 (.012)	10:1 (.012)	12:0 (<.001)	9:3 (.146)
A2, edit (judged clean fix)	10:7 (.629)	9:5 (.424)	12:1 (.003)	4:9 (.267)

6.4 RQ4: On real pull requests, the skill avoids harm but fixes little

Real descriptions were much harder (Figure 5). No condition scored above 0.3 for addressing the reviewer’s critique. Twelve of the 21 critiques depended on something outside the text, such as the diff or a CI run; on the 9 that could be judged from the text alone the best score was still under 0.39.

Without an instruction the models addressed critiques about as often as with v0.1.0 (0.27–0.28 against 0.28–0.29), but 21–38% of their outputs added a fact and 23–27% damaged needed text. Changes that both judges flagged include narrowing an access rule from “other” DAGs to “unauthorized” ones, deleting an SPDX license header, and editing the author’s AI disclosure. Seven A2 descriptions carry Airflow’s **Generated-by:** disclosure line. Across their 70 outputs per condition, that line was altered or deleted 10 times without an instruction, 9 times with one sentence, once each with three sentences and v0.1.0, and never with v0.2.0. GPT models added ChatGPT or Codex to another author’s disclosure in 8 outputs, and GPT-6 Luna deleted the line in 5.

With v0.2.0, outputs almost never added a fact (0–1%) or damaged needed text (1–4%), but the skill addressed fewer critiques (0.19) and left 24% of descriptions unchanged. On clean fixes it scored 0.18–0.19: above no instruction (0.12–0.15) and the three-sentence instruction (0.07–0.10),

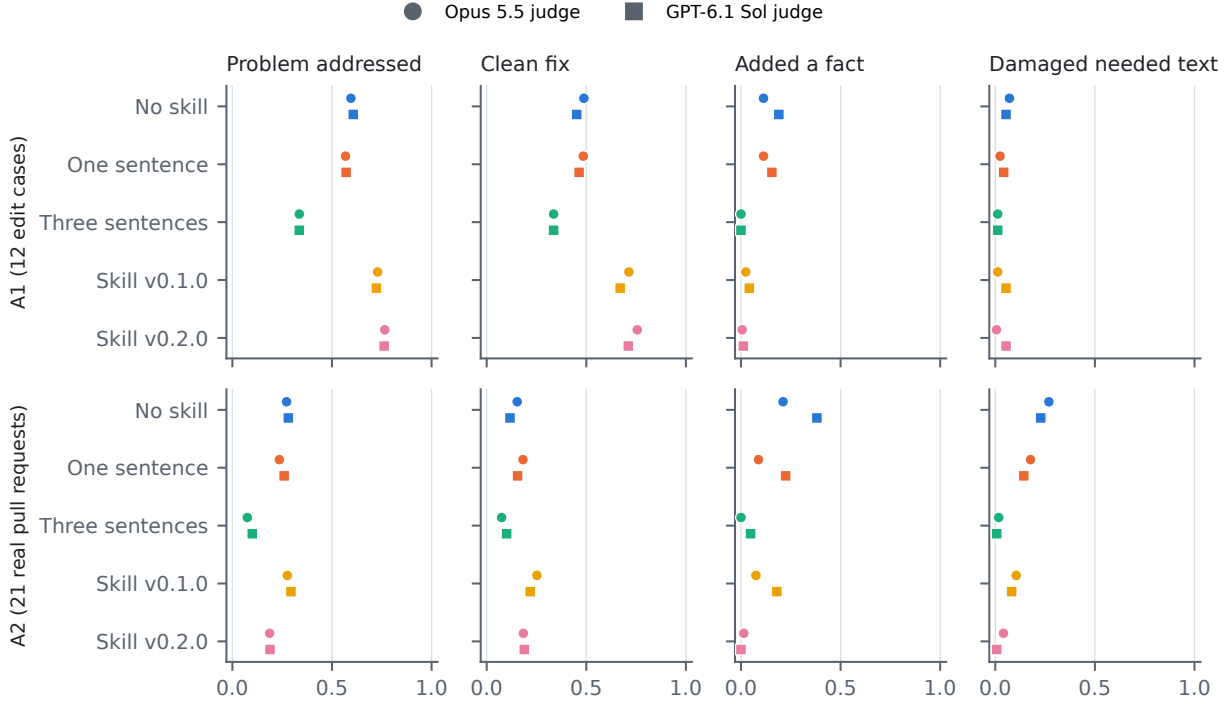


Figure 5: Judged edit cases in A1 and A2. A clean fix counts the addressed score only when the revision neither added a fact nor damaged needed text.

and below v0.1.0 (0.22–0.25). Only the difference from the three-sentence instruction is significant (12:1, $p = 0.003$; against no instruction 10:7, $p = 0.63$; against v0.1.0 4:9, $p = 0.27$). By critique type, v0.2.0 fell furthest behind on misleading titles (0.13 against 0.38 without an instruction, 4 cases) and on text reviewers called AI-sounding (0.07 against 0.21, 2 cases): style problems that v0.2.0 stopped targeting. With so few cases, read this as a direction to check, not a result.

6.5 Judge agreement

The two judges agreed closely on whether an edit addressed the problem (exact agreement 91%, weighted $\kappa = 0.93$, 1,650 outputs) and on whether a change to a keep case was harmful (88.5%, $\kappa = 0.73$, 667 outputs). They agreed less on added facts ($\kappa = 0.59$; the Sol judge flagged 14% of outputs and the Opus judge 7%) and on damage ($\kappa = 0.46$), so Table 4 reports both. Neither judge favored its own model by much. The Opus 5.5 judge scored Opus 5.5 outputs 0.018 higher than the Sol judge did, against -0.011 for other models’ outputs; the Sol judge scored Sol outputs 0.021 higher than the Opus judge did, against 0.003 for the rest. These gaps are small next to the differences between conditions. A first pass on A1 that hid the editor’s note from the judges gave the same order: 0.73 for v0.2.0, 0.30–0.32 for the three-sentence instruction, and 0.58–0.59 without an instruction.

6.6 RQ5: Cost

Loading the skill costs time and tokens on every task that uses it (Table 5). In Claude Code, a v0.2.0 run cost 2.5 to 3.3 times as much as a run without it (Opus 5.5: \$0.123 against \$0.044) and took 3–9 seconds longer. In Codex, input tokens rose from about 11–12k to 27–28k and run time

Table 4: Judged edit cases. In each pair the first number is from the Opus 5.5 judge and the second from the GPT-6.1 Sol judge. *Addressed* and *clean fix* are mean scores from 0 to 1; a clean fix counts the addressed score only when the revision neither added a fact nor damaged needed text. *Added a fact*, *damaged*, and *unchanged* are percentages of outputs. Models weigh equally; Claude runs are averaged first.

	<u>Addressed</u>		<u>Clean fix</u>		<u>Added a fact</u>		<u>Damaged</u>		Unchanged
<i>A1 (12 cases)</i>									
No skill	0.59	0.61	0.49	0.45	11	19	7	5	0
One sentence	0.57	0.57	0.48	0.46	11	16	2	4	15
Three sentences	0.34	0.34	0.34	0.34	0	0	1	1	56
Skill v0.1.0	0.73	0.72	0.71	0.67	2	4	1	5	9
Skill v0.2.0	0.77	0.76	0.76	0.71	1	1	1	5	12
<i>A2 (21 real pull requests)</i>									
No skill	0.27	0.28	0.15	0.12	21	38	27	23	0
One sentence	0.24	0.26	0.18	0.15	9	22	18	14	7
Three sentences	0.07	0.10	0.07	0.10	0	5	2	1	13
Skill v0.1.0	0.28	0.29	0.25	0.22	8	18	10	8	4
Skill v0.2.0	0.19	0.19	0.18	0.19	1	0	4	1	24

Table 5: Mean seconds and cost per run on the held-out sets. Claude cost is the API-equivalent estimate that Claude Code reports, not a bill; for Codex the table gives input tokens.

	No skill		Three sentences		Skill v0.2.0	
	s	cost	s	cost	s	cost
Opus 5.5	14	\$0.044	12	\$0.039	17	\$0.123
Opus 4.6	13	\$0.038	18	\$0.044	22	\$0.093
Sonnet 5.5	9	\$0.018	8	\$0.018	12	\$0.059
GPT-6.1 Sol	15	12.0k	18	12.0k	23	28.4k
GPT-6 Astra (high)	12	11.9k	14	11.9k	19	28.3k
GPT-6 Astra (low)	11	11.9k	13	11.9k	18	28.3k
GPT-6 Luna (max)	16	11.3k	22	11.3k	38	27.0k

from 11–16 to 18–38 seconds. Version 0.1.0 used more tokens than v0.2.0 in every configuration. The short instructions cost nothing measurable.

7 Discussion

What the skill adds. For leaving clear text alone, not much beyond a short instruction. Three sentences in a prompt got most configurations to 79–100% on keep cases at no extra cost. The skill’s measurable contribution is on text that does need an edit. A short instruction to edit “only where there is a concrete problem” does not say what a problem is, and the models read it as permission to leave unsupported claims in place. The skill names the problems that reviewers in Section 2 raise most, and on held-out text it fixed them cleanly about twice as often as the three-sentence instruction did.

When to use which.

- If an agent mostly touches text that is already fine, such as tidying or translating documentation, put the three-sentence instruction in `AGENTS.md` or `CLAUDE.md`.

- If it drafts or revises pull request descriptions, release notes, or review replies, where overclaims are likely, the skill catches more of them at 2.5 to 3.3 times the cost per run.
- With either, read the output. GPT-6 Luna at max effort ignored the skill’s first rule on about half of the keep cases.

What the skill does not do. It does not make text sound human and cannot supply evidence the author did not give. In A2, most of what reviewers objected to needed facts only the author had: the diff, the CI run, the benchmark. No condition cleanly fixed more than a quarter of those critiques. What v0.2.0 did reliably on real descriptions was not make them worse, including leaving other people’s disclosure lines alone. It also fixed fewer problems than v0.1.0, mostly misleading titles and text reviewers called AI-sounding. That is the cost of its narrower scope and the first place to look in a next version.

A lesson from v0.1.0. The longer first version, built around lists of patterns to fix, made models edit more, not better. For the GPT configurations it left fewer clear texts alone than the one-sentence instruction did, and it used more tokens than v0.2.0 everywhere. Rules about when not to edit did more than lists of what to fix.

8 Threats to validity

- **Who wrote the cases.** The development set was written by the maintainer with an AI assistant, and the skill was tuned on it. A1 was drafted by Claude Opus 5.5 after release and shares its authors’ idea of a good edit; the literal checks for A1 edit cases encode that idea. A2 and the judges are the checks against this, and A2 is small.
- **Scope of A2.** Twenty-one English pull request descriptions from four projects, found by phrase search. They do not cover documentation, comments, or other languages.
- **Literal scoring** misses paraphrases that keep the meaning and counts a harmless fix to clear text as a failure. The judged harm rate on keep cases separates those.
- **Judges.** Both judges are tested models and share training-era habits with the editors they grade. Labels were blind to model and condition.
- **Sample size.** Claude configurations ran twice and Codex configurations once, so per-model results are noisy, and A1 and A2 have 24 and 21 cases. The pooled tests use cases as the unit and treat the seven configurations as fixed.
- **Harness asymmetry.** Task prompts, output format, and skill activation differ between Claude Code and Codex. Claude outputs could carry a note to the author and Codex outputs could not, which favors Claude where the right move is to ask. Within-configuration comparisons are unaffected; cross-vendor comparisons are not meaningful.
- **Baselines.** The short instructions are in English and were not tuned. A better short instruction may close part of the gap reported here.
- **Earlier runs.** An earlier report on GPT-6 Astra used a sandbox in which the agent could not run shell commands, so it never read the skill’s reference files. Those runs are not used here; all Codex skill runs in this report opened `SKILL.md`.

9 Conclusion

Current models rewrite clear text by default, and much of that rewriting changes meaning. A three-sentence instruction prevents most of it at no cost. The polish-open-source-prose skill adds

two things the short instruction does not: it fixes unsupported claims instead of leaving them in place, and on real pull request descriptions it almost never adds a fact or alters a disclosure. It does not fix most of what reviewers object to, because most of that needs facts only the author has.

For a contributor the practical advice is short. Tell the agent to leave clear text alone and not to add facts. Use the skill when the agent writes claims about your change. Then check what it wrote against what you did.

Use of AI tools

The experiments were carried out with Claude Code running Claude Opus 5.5, under the author’s direction. It wrote the scoring, judging, and analysis scripts, drafted the A1 cases, labeled the sample of review threads, ran the models, and drafted this report. The author chose the research questions, conditions, models, and data sources, approved each step before it ran, and takes responsibility for the content. Claude Opus 5.5 is also one of the tested models and one of the two judges; Sections 6.5 and 8 discuss what that may bias.

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A Reproducing

Raw runs, event logs, and the A2 cases live under \$POLISH_EVAL_DATA (see evals/README.md). From the repository root:

```
POLISH_EVAL_DATA=/mnt/d/Data python3 evals/score_benchmark.py
POLISH_EVAL_DATA=/mnt/d/Data python3 evals/judge_outputs.py --judge both
POLISH_EVAL_DATA=/mnt/d/Data .venv/bin/python evals/analyze_benchmark.py
tectonic docs/report/report.tex
```

The judge script calls the `claude` and `codex` command-line tools; set `CLAUDE_BIN`, `CODEX_BIN`, and `CODEX_AUTH` if they are not on the path. `evals/results/benchmark.json` holds every scored output for the development set and A1, and `evals/results/judgments.json` every judge label for them. A2 outputs and labels stay with the A2 cases. `evals/results/analysis.json` holds the aggregates in this report, including A2.

B Examples from held-out set A1

Outputs are verbatim from the first run. Both cases are English pull request text. Claude Code outputs also carried a note to the author, summarized here.

An unsupported speedup. Input: “This avoids copying each batch twice in `SortExec`, which makes queries 3x faster.” The rubric: the speedup has no benchmark, data size, or command; keep the mechanism, and drop the number or ask where it came from.

- Opus 5.5, no skill: “This change stops `SortExec` from copying each batch twice. On [benchmark or query set], queries ran about 3x faster.” The note asks the author to fill in the placeholder, which sits inside the text meant to be pasted.
- Opus 5.5, three sentences: unchanged. The note calls the sentence “already clear and correct” and suggests naming the benchmark.
- Opus 5.5, v0.2.0: “This avoids copying each batch twice in `SortExec`.” The note explains the removal and asks what was measured, the numbers, and which queries speed up.
- GPT-6.1 Sol, v0.2.0: unchanged, with no note. The skill does not always work.

A test result stretched into a conclusion. Input: “I ran `pytest tests/test_xcom.py` and everything passes, so the change is fully tested and safe to merge.” The rubric: one test file passing does not make the change fully tested or safe to merge; keep the command and its result and drop the conclusion.

- Opus 5.5, no skill: “I ran `pytest tests/test_xcom.py` locally, and all tests pass.” Both judges flagged “locally” as a fact the author did not give.
- Opus 5.5, three sentences: unchanged. The note says that one test file passing does not prove the change is “fully tested and safe to merge”, but leaves the sentence as it was because it “is clear and grammatical”. The model saw the problem and kept it, as the instruction to leave clear text unchanged allowed.
- Opus 5.5, v0.2.0: “I ran `pytest tests/test_xcom.py` and all tests in that file pass.” The note asks which commit the tests ran on, whether other tests ran, and whether CI passed. Both judges marked this as addressed, but it fails the literal check because “all tests” is on the case’s must-not-introduce list: literal scoring counted a good edit as a failure.